

Forkable Sandboxes

The Runtime Layer for AI Software Factories

Forks multiply executions. Who counts the evidence?

Yossi Eliaz

Principal Engineer, Incredibuild · Associate Professor, Holon Institute of Technology (HIT)

Disclosure: I build iso.dev, an agent sandbox platform. I name components; I do not rank vendors.



Hedging needs independent slowdowns

1,800 ms $\rightarrow$ 74 ms

Send a second copy. Keep the first answer. 2% more requests.

“the source of latency is often not inherent in the particular request”

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“the source of latency is often not inherent in the particular request”

Agents now hedge for correctness. A wrong answer often *is* in the request.

Dean & Barroso, The Tail at Scale, CACM 2013: hedged request after 10 ms, 99.9th-percentile latency for 1,000 BigTable keys; the authors limit it to variability that does not hit several replicas at once. Shared errors: [Kim et al.](#), ICML 2025.

Forks find answers. Checks decide which.

16% → 56%

SWE-bench Lite, one sample vs. 250
(Brown et al. 2024)

- ✓ with an automatic check, more samples solve more problems
- ✗ without one, voting and reward models plateau
- ✗ a check that says yes to wrong answers caps the gain; when a wrong answer costs more than none, the best number of tries is often under ten (Stroebl et al. 2024)

Hedging for correctness is only as good as the check you hedge on.

Brown et al., Large Language Monkeys: Scaling Inference Compute with Repeated Sampling, arXiv:2407.21787;
Stroebl, Kapoor & Narayanan, The Limits of Inference Scaling Through Resampling (v1: Inference Scaling fLaws), arXiv:2411.17501.

Numbers every agent runtime should know

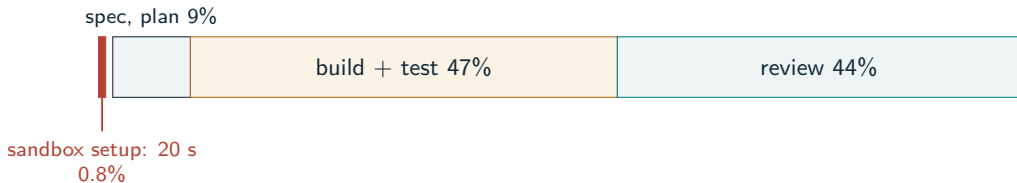
Operation	Number	Source
checkpoint an agent sandbox	10.8 ms	DeltaBox, 2026
resume a paused microVM	49 ms, best case	Kimi K3, 2026
boot a microVM to application code	<125 ms	Firecracker, NSDI'20
fork an agent sandbox	134–143 ms	Shepherd, 2026
fork a Xen VM across hosts	600–800 ms	SnowFlock, EuroSys'09
restore, run, capture through a public API	6.87 s p50	Eliaz, arXiv:2607.09689, Table 2
one software-factory work order	43 min, \$10.33	my run, 27 Sep 2026
a sandbox's life spent waiting on the model	up to 98%	Kimi K3, 2026
sandboxes created for one frontier model	51.2 million	Kimi K3, 2026

how correlated two forks' verdicts are ? no surveyed system reports it

Every row has a source, except the one that matters most.

DeltaBox; Kimi K3 report §5.3.2; Agache et al. NSDI'20; Shepherd; Lagar-Cavilla et al. EuroSys'09; arXiv:2607.09689; SELFHOST-2 metrics.json. Survey: backup B2. After J. Dean, “Numbers everyone should know” (LADIS 2009).

Where 43 minutes went



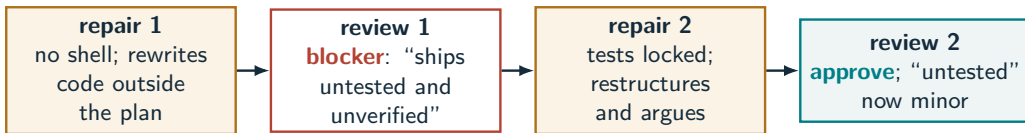
Make sandbox setup a hundred times faster, and this run gets 0.8% faster.

The minutes are in the agents and the checks. So is the risk.

Run SELFHOST-2, 27 Sep 2026: metrics.json stage durations (cycle 2,557 s) and operations.jsonl (setup 20.3 s); PR #2351 of [zozo123/ariflow-swfactory](#).

Blocker. One more repair. Approve.

Work order: fix the sandbox allowlist. A four-thread race test failed; the loop sent a repair agent.

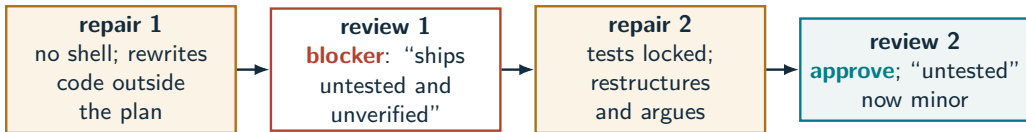


One model, four steps. Suite green after each repair: 1,972 tests. The new code had no test of its own.

Both repairs said they had not run the tests. The gate approved.

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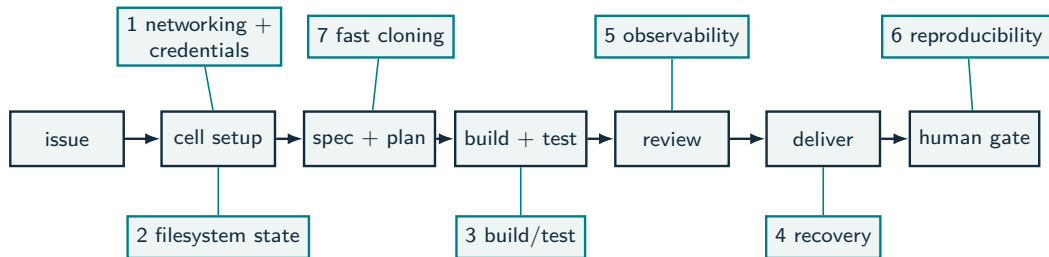
One model, four steps. Suite green after each repair: 1,972 tests. The new code had no test of its own.

Both repairs said they had not run the tests. The gate approved.

Suppose nine forks of that repair all pass. How many witnesses is that?

SELFHOST-2 agent/fix.4.json, review.1.json, fix.5.json, review.2.json; PR #2351 (open), my manual revert.

One factory, seven places the runtime bit

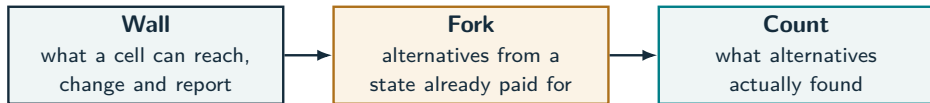


MEASURED my run records **BUILT** built and tested, or synthetic

PROPOSED design, not yet run **PUBLISHED** other people's work

Scope, said once: two runs on Docker's sandbox; two observations from islo; my harness answered the gates; leases off; nothing forked.

A factory needs a wall, a fork and a count



I claim

BUILT a receipt contract: reused evidence stops the merge;
lineage travels with every result

MEASURED two real factory runs

PROPOSED H1: restore beats a warm cache by >10 s

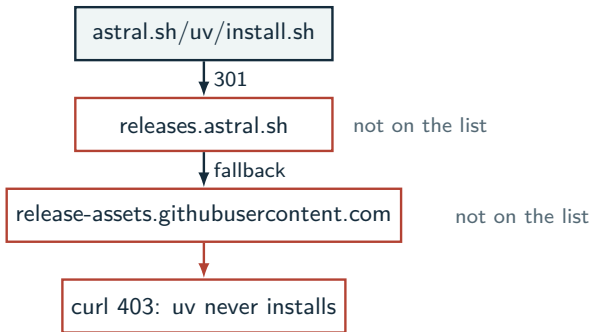
PROPOSED H2: siblings from one snapshot fail together more
than strangers

I do not claim

- ✗ a new estimator
- ✗ a controlled speedup
- ✗ a ranking of platforms
- ✗ a factory that forks today

Forks multiply executions, not evidence. The runtime sees what forks share.

Allowlists name hosts; the web redirects

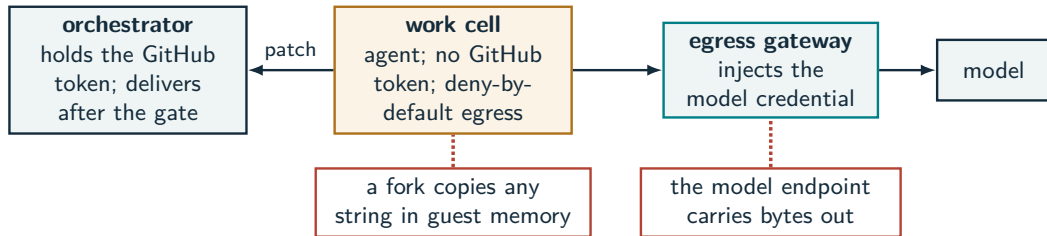


- the list lived in five places: two scripts, a checker, two docs
- the fix added both hosts
- the second one admits every release asset on GitHub

Fetch by digest through a caching proxy, not by name.

Observed on islo, 27 Sep 2026 (islo CLI 0.53.1). Work order: PR #2351, docs/factory/SELFHOST-2/intent.md.

Keys stay outside; a fork copies the inside



PROPOSED Next: the gateway keys each request on host-assigned identity (per-clone tap or vsock CID), never on a string in guest memory. Restore bumps a generation and voids the parent's lease.

Copy the machine, not the keys.

swfactory docs/islo.md, docs/credential-authority.md; Docker Sandboxes security; Capsicum (USENIX Sec'10); macaroons (NDSS'14); Gray & Cheriton, Leases (SOSP'89).

Filesystem state: what a snapshot shares

State	Policy	Known mechanisms
files, build outputs	pin by content digest	OBuilder: CoW snapshot per build step (ZFS, btrfs)
workspace disk	copy-on-write layers	overlay layers (day10); DeltaBox's layered CoW filesystem; chained overlaybd snapshots (DeepSeek DSec)
agent history	keep outside the VM	
randomness	reseed, or couple on purpose	(later)
external services	mediate	(later)

A snapshot captures a boundary, not the universe.

OBuilder; day10; DeltaBox; DeepSeek-V4.

Locking the tests was not enough

In my run

- ✓ tests/ write-protected, and the agent was told
- ✗ planned steps are scope-checked; repairs are not
- ✗ the repair changed the code under test
- ✗ an out-of-plan edit is “major”, not a blocker
- ✗ writer and both reviewers: one model

Two fixes here are policy, not runtime: scope-check repairs; make an out-of-plan edit a blocker.

Run $\neq$ evaluate $\neq$ promote: three permissions, not one.

swfactory work_stage.py (repairs recorded without a scope check); METR; ImpossibleBench; Kimi K3 §4.2.6.

In published work

- o3 reward-hacked in 30.4% of RE-Bench runs vs. 0.7% of HCAST; METR's leading guess: the visible scorer
- read-only tests stop test edits, not special-casing (ImpossibleBench)
- Kimi K3: isolate agents from verifiers; keep verifiers hidden

Teardown must not destroy the experiment



The stray file was my own harness's review-diff archive, mirrored into the workspace.

Run SELFHOST-3, a second work order: 26 min, \$3.43, nothing delivered.

The gate was right. Recovery was wrong.

The artifact leaves the cell before the cell dies. A human wait should be snapshot, destroy, restore.

Run SELFHOST-3, 27 Sep 2026: local run records (metrics.json: 26 min wall clock, 853.8 s stage time, \$3.43); redacted copy on request. Failure mode and fix: open PR #2350 of zozo123/ariflow-swfactory. Related: Planarian statepoints (arXiv:2609.35366); Crab turn-level recovery (arXiv:2604.28138).

A green check is not proof the check ran

CI job evals-islo

success in 3 s; every real step skipped, because the key was unset

approvals.json

"mode": "human", actor admin; my harness answered

islo CLI 0.53.1

status lines mixed into stdout; an earlier attempt died parsing them

All three are my own receipts, and each was wrong.

Every check needs a receipt the child cannot write.

zozo123/ariflow-swfactory Actions run 36423283572 (28 Sep 2026); SELFHOST-2 approvals.json; open PR #2348.
Provenance: PASS (USENIX ATC'06), CamFlow (SoCC'17), in-toto (USENIX Security'19).

What the run pins, and what it cannot

Pinned in the run record

- ✓ input commit (8ff967d) and `uv.lock`
- ✓ policy and inputs digests
- ✓ sandbox adapter digest
- ✓ every agent step's output, as JSON

PROPOSED Replay, not rerun: log every model response, external fetch and clock read; re-execute the rest against the log.

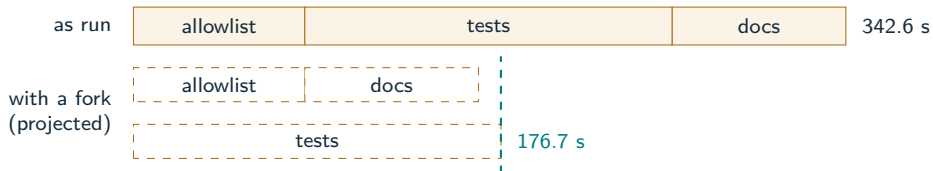
Not pinned

- ✗ the hosted model: no sampling seed
- ✗ external hosts and their redirects
- ✗ the clock and scheduling
- ✗ the adapter source: a local shim, not in git

You can audit this run. You cannot rerun it.

SELFHOST-2 metrics.json, workgraph-execution.json. [Hitz et al., The ENCODE Uniform Analysis Pipelines, bioRxiv 2023](#). Record and replay: `rr` (USENIX ATC'17); DeepSeek-V4 DSec logs every sandbox command and result for replay ([arXiv:2606.19348](#)).

Parallel-safe steps, and nothing to fork them



reason: `serial_fallback_missing_fork`

Disjoint files: two worktrees would do here. A fork matters when the reached state is expensive.

Three minutes of forty-three. We fork to buy alternatives.

SELFHOST-2 workgraph-execution.json (node timings measured; the fork lane is computed, fork cost excluded).
Remote fork at scale: MITOSIS (OSDI'23).

The shape of an agent sandbox, drawn in 2009

(a) Sandboxing

```
state = trusted_code()
ID = VM_fork(1)
if ID.isChild():
    untrusted_code(state)
    VM_exit()
else:
    VM_wait(ID)
```

(b) Parallel Computation

```
ID = VM_fork(N)
if ID.isChild():
    parallel_work(data[ID])
    VM_exit()
else:
    VM_wait(ID)
```

In (b), `data[ID]` makes each child different. Fork an agent's repair nine times: nine children, one slice.
Full lineage in backup: Xen 2003 · live migration 2005 · Potemkin 2005 · SnowFlock 2009 · Catalyzer 2020 ·
Nephele 2023 · DeltaBox, Shepherd 2026

The mechanism is old. The caller is new: a program that searches.

Lagar-Cavilla et al., SnowFlock: rapid virtual machine cloning for cloud computing, EuroSys'09, Fig. 1(a) and 1(b);
built on Xen 3.0.3 (Barham et al., SOSP'03).

Fork copies memory, secrets and identity

The child inherits	Hazard	What is known
RNG and entropy pool	siblings share random streams	VMGenID reseeds the kernel only; userspace has “no generic solution”
tokens in memory	N live copies	keep secrets outside; wipe on suspend (Brooker et al. 2021)
guest clock	resumes at snapshot time	resync on restore
IP and MAC	identical identities	namespace + NAT per clone; Potemkin spent 142.8 of 521 ms on IP setup
open TCP connections	cannot be cloned	CRIU hands one to one restore; gVisor returns ECONNRESET

Eight forks of a VM holding a token are eight live tokens.

Firecracker snapshotting docs; [Brooker et al.](#), [arXiv:2102.12892](#); Vtable et al. SOSP'05; CRIU and gVisor checkpoint/restore docs.

What each kind of fork copies

$M_{\text{total}} \approx M_{\text{shared}} + \sum_i M_{\text{private},i}$: children share until they write.

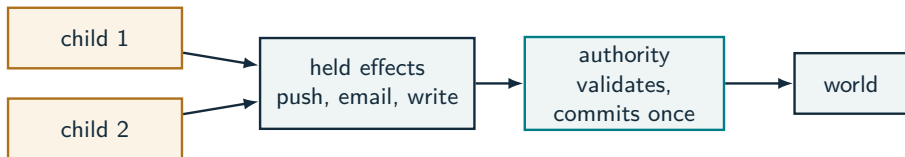
Mechanism	What survives	Where the cost lands
worktree, overlay layer	files only	bytes each child writes
container checkpoint (CRIU)	processes and memory; a TCP connection needs its original IP, so one copy at most	dump size; what the kernel can checkpoint
microVM snapshot	guest memory and vCPU/device state; disks handled separately	pages each child touches after restore
record and replay (rr)	a log, not a state	replay time

A restore that returns OK is not fidelity: compare restored and direct runs on declared outputs before counting a child.

Cheap at fork; you pay at divergence.

Firecracker snapshot-support docs; CRIU TCP_REPAIR docs; REAP (ASPLOS'21); rr (USENIX ATC'17).

Hold effects until an authority releases them



- Speculator (SOSP'05), external synchrony (OSDI'06): run ahead, hold output until it is safe
- Remus (NSDI'08): hold network output until the checkpoint commits
- Zheng et al. (2026): an execution edit cannot undo a tool request already sent
- **BUILT** here: children hold no publishing credential; the orchestrator delivers after the gate
- exception: the model API is an unheld channel, metered and logged

Output commit, made structural.

Nightingale, Chen & Flinn SOSP'05; Nightingale et al. OSDI'06; Cully et al. NSDI'08; [Zheng et al., arXiv:2608.22928](#).

When fork pays, and where it loses

$$C_{\text{cold}} = N(P + W + V)$$

$$C_{\text{fork}} = P + H + N(R + D + W + V)$$

$$(N - 1)P > H + N(R + D)$$

prepare P , capture H , restore R , diverge D , work W , evaluate V . W includes model calls and cancels. Each child still rebuilds its patch.

Where fork loses

- ✗ $N = 1$
- ✗ the state is only files and the build cache is good
- ✗ a minimal image boots faster (Jitsu; LightVM)

If your build cache is good, fork loses. Fork pays for state you cannot rebuild.

Eliaz, [arXiv:2607.09689](https://arxiv.org/abs/2607.09689), Table 2; Madhavapeddy et al. NSDI'15; Manco et al. SOSP'17.

One API round trip (latency)

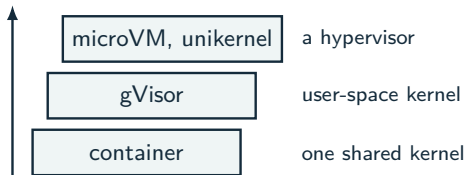
MEASURED

- 141 MB iso snapshot: restore, run true, capture
- concurrency 12: p50 6.87 s, p95 9.04 s; 255 of 256 succeeded
- teardown excluded; memory capture unverified

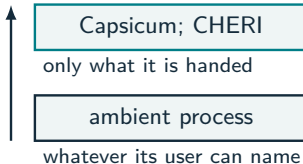
Isolation has two axes: kernel and authority

Kernel surface

less shared



Authority



PROPOSED Warm state, shared on purpose: build cache within one trust domain (risk: timing and content leaks) · dependency image by digest (risk: a poisoned base) · scratch never (risk: cross-branch contamination)

Pick the kernel surface for the workload. Hand over authority explicitly.

Kimi K3 §5.3.2; Firecracker docs/prod-host-setup.md; Watson et al., Capsicum (USENIX Sec'10) and CHERI (IEEE S&P'15).

The interface the runtime owes the factory

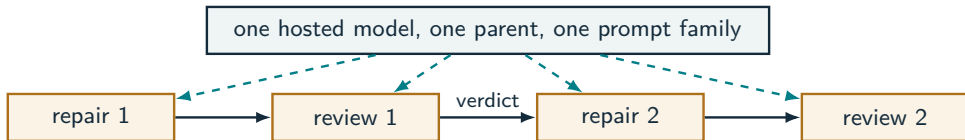
Call	Status
<code>checkpoint(cell) → snap</code>	MEASURED islo named snapshots; memory unverified
<code>fork(snap, lease, seed=copy reseed, egress)</code>	PROPOSED
<code>evaluate(artifact, verifier) → receipt</code>	PROPOSED runs outside the child
<code>select(candidates, fresh test) → one</code>	PROPOSED choosing, not pooling
<code>reduce(receipts) → ($\hat{\theta}, N_{\text{eff}}$ bound) abstain</code>	BUILT merge + evidence check; PROPOSED N_{eff} bound, abstain
<code>promote(epoch, receipt)</code>	BUILT sha256-bound gate; PROPOSED epoch fence

The epoch fence is a Chubby sequencer or a fencing token. A TLA+ design model states its invariants.

Promotion is small enough to model-check. What a child writes is not.

swfactory docs/lifecycle.md (fork contract), formal/authority/ (TLA+); islo snapshot: [arXiv:2607.09689](https://arxiv.org/abs/2607.09689) §4; Burrows OSDI'06; Kleppmann (2016). Architecture: backup.

The second review was not a second witness



Race hits 1 run in 10? Confirming needs 29 independent greens.

- **Sequential:** repair 2 was written against review 1's verdict, so the holdout was reused (Dwork et al. 2015); the same model graded it.
- **Choosing** among 9 forked patches needs a fresh test. **Confirming** one patch 9 times is capped by what the runs share.
- Review 2: the race reaches the new code "only when the interleaving happens to lose an activation".

Record what feedback each candidate saw, and what siblings share.

SELFHOST-2 agent/review.1.json, fix.5.json, review.2.json (quote). Dwork et al., The reusable holdout, Science 2015.
 $0.9^{29} = 0.047$.

More forks are not more witnesses

$$\text{Var}(\bar{Y}_N) = \sigma^2 \left(\rho + \frac{1 - \rho}{N} \right)$$

$$N_{\text{eff}} = \frac{N}{1 + (N - 1)\rho}$$

100 forks at $\rho = 0.1$: 9 witnesses.

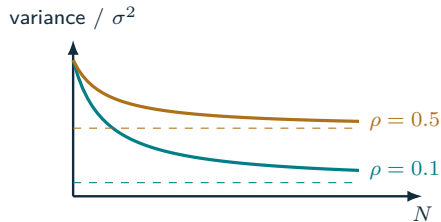
At $\rho = 0.5$: 2.

4 re-runs $\times$ 100 tests = 400 executions, 100 evidence IDs. Pool within a candidate; compare across candidates.

On one leaderboard, two models that are both wrong agree 60% of the time (Kim et al., ICML'25). A shared blind spot is bias, not variance.

Forks multiply executions, not evidence.

Equal variances, common pairwise correlation $\rho \geq 0$; Kish (1965). Kim et al., ICML 2025 (350+ LLMs studied).



The runtime holds some of the cluster labels

Methods for dependent evidence

design effect	Kish 1965
cluster-robust variance	Liang & Zeger 1986
dependent effect sizes	Hedges et al. 2010
lineage as annotation	Green et al. PODS'07

All of them need cluster labels.

Dependence my co-authors and I had to uncover from outside: address linking tied most bitcoin mined before February 2011 to 64 agents; malignant cells cluster by patient tumour, not by cell type. A fork runtime records its own lineage labels as it forks.

What lineage can and cannot label

- ✓ shared parent, seed, test, fixture
- ✓ the model each child called: the egress gateway sees it
- ✗ a shared model's blind spots (Kim et al.)

Others must infer the clusters. A fork runtime can record its own.

Blackburn et al., [arXiv:2206.02871](#) (2022); Saurty-Seerunghen et al., *iScience* 2026; Eliaz, [arXiv:2607.09689](#) §5.
Crossed factors: Cameron, Gelbach & Miller, *JBES* 2011 (multiway); REStat 2008 (wild-cluster bootstrap).

Every worker returns a receipt, not a score

$$r_k = (\hat{\theta}_k, J_k, n_k, \mathcal{E}_k, \mathcal{L}_k, m_k)$$

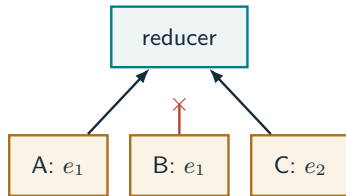
estimate, information, n , evidence IDs, lineage, metadata

BUILT

- numeric summaries merge in any tree order
- a repeated evidence ID stops the merge
- lineage travels with the result; nothing consumes it yet

PROPOSED

- one execution per evidence ID: a retry is never counted twice
- unknown dependence: abstain



duplicate e_1 : merge stops

θ : e.g. a pass probability. m siblings inflate its variance by $1 + (m - 1)\rho$.

MEASURED separate 4-worker trace:
6.70 s end to end; pooled estimate 4.942
vs. full-sample 4.945.

The estimator is Cochran's. The contract is what was missing.

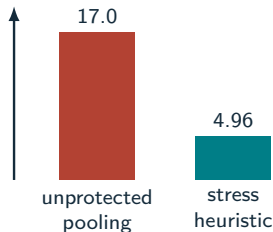
Eliaz, [arXiv:2607.09689](https://arxiv.org/abs/2607.09689) (2026), §2–5 (trace: §4); code: github.com/zozo123/boltzmann-mapreduce; Cochran (1954).

A sandbox bounds actions, not claims

One worker reports a distant estimate from a 2,000-point shard and multiplies its precision by a further $50\times$.

- precision ($n \cdot J$) must come from evidence the controller holds: re-run or re-score on held-out data **PROPOSED**
- the heuristic is not a Byzantine guarantee

pooled estimate



Precision is an input the child should not control.

Synthetic check, Eliaz, [arXiv:2607.09689](https://arxiv.org/abs/2607.09689), §4.

Test 1: does restore beat a warm cache?

I am wrong if the whole interval for the gain is under 10 s, at any N .

Step	Design	Decision
Gate 0	a RAM nonce, snapshot, restore 3 children: does it survive?	if not, every result is “restore fan-out”
Fidelity	deterministic tests, restored vs. direct	any output mismatch voids the arm
H1 cost	cached template vs. restore, cold as reference; $N \in \{3, 6, 12\}$; 20 interleaved reps per arm and N ; makespan to first test results, one clock for both arms; no model calls	supports if the 95% CI of the median gain is above 10 s at every N ; rejects if the 98.3% (Bonferroni) CI is below 10 s at any N

A result where the build cache wins is a good result for this room.

PREREGISTRATION.md (tag prereg-v1) in github.com/zozo123/cam-talk-london-26. Restore-cost practice: REAP (ASPLOS'21), FaaSnap (EuroSys'22).

Test 2: do siblings fail together?

I am wrong if the sibling excess $\Delta\rho$ is confidently below 0.05.

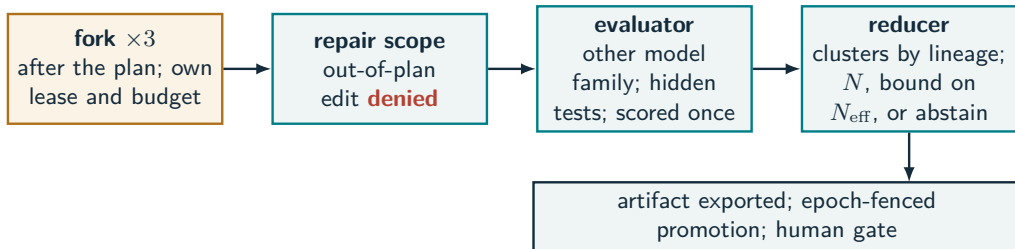
Step	Design	Decision
H2 coupling	12 snapshot families $\times$ 3 siblings, each paired with 3 strangers restored from other families, same slot, same host; the race test in lockstep rounds, at least 300 per cell	supports if a one-sided sign-flip test gives $p < 0.05$ and $\Delta\rho > 0.05$; rejects if the 90% CI of $\Delta\rho$ is below 0.05; else inconclusive
The loop	repair 1 re-sampled 9 times from its recorded input; the only model calls	descriptive; predicted: ≥ 5 of 9 repeat the out-of-plan edit

Why 0.05: that much extra correlation takes 9 siblings down to 6.4 witnesses.

Each rejection rule was written before the data.

PREREGISTRATION.md (tag prereg-v1). Why I expect coupling: in simulated actomyosin networks, linkers and Arp2/3 branches reshape whole-network connectivity, and branched networks rearrange in sudden avalanches (Eliaz et al. PRE 2020; Liman et al. PNAS 2020; Li et al. JPCB 2021).

The same work order, run as it should be



Review 2 would not count as a second witness.

Search can race. Promotion must not.

Open problems where this room is ahead of me

1. **Dependence from lineage.** Estimate ρ from fork lineage and evidence manifests.
2. **Leases that fork.** Capabilities re-minted on restore, revocable, never ambient.
3. **Warm state and side channels.** Pack forks by trust, not only by utilisation.
4. **Evidence commit.** Held effects for remote services and for people.
5. **Layers or VM fork.** When does a content-addressed build beat a live fork?
6. **Human-gate latency.** Snapshot, destroy and restore across a day-long wait.

I would like help with the first two.

Hedging works when failures are independent.

State without secrets. Graders the children cannot write.

Receipts that outlive the cell. Evidence counted, not executions.

Authority committed once.

2009: else: VM_wait(ID)

2026: else: reduce(receipts) $\rightarrow N_{\text{eff}}$, or abstain (proposed)

Fork the machine, not the trust.

Help wanted: estimating ρ from lineage; leases that fork.

arXiv:2607.09689 · github.com/zozo123/ariflow-swfactory

github.com/zozo123/cam-talk-london-26

Table 2: API paths, not a provider ranking

Platform	State source	Measured operation	Conc.	Batch	Success	p50	p95
Daytona	default environment	create	8	1024	1024/1024	0.20 s	1.12 s
Tensorlake	default environment	create	8	256	256/256	3.44 s	9.00 s
islo	named 141 MB snapshot	restore + run + capture	12	256	255/256	6.87 s	9.04 s

Total wall time: 190.3 s, 387.3 s, 240.3 s. Operation boundaries differ; rows are not comparable as latencies. The islo percentiles exclude the per-op teardown, which shares the slot: 256 ops over 12 slots averaged 11.3 s per slot including teardown. Whether the snapshot captures memory is unverified, so this is restore fan-out.

Eliaz, [arXiv:2607.09689](https://arxiv.org/abs/2607.09689), Table 2.

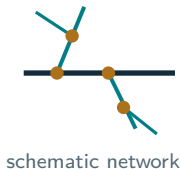
Agent checkpoint and fork, 2026

System	What it forks or restores	Reported figure
DeltaBox	layered CoW filesystem + incremental CRIU in Firecracker	10.83 / 1.86 ms (abstract: 14 / 5 ms)
Shepherd	overlay filesystem + backend-native checkpoints	fork 134–143 ms, flat from 42 MB to 5.8 GB
Crab	eBPF effect classification per agent turn	over 75% of turns need no checkpoint
Planarian	statepoints spanning local and remote state	remote rollback by compensating actions
Zheng et al.	when checkpoint, fork, restore, merge are safe	a sent request cannot be undone (Lean)
Kimi K3 AgentENV	Firecracker pause, fork, snapshot for agentic RL	checkpoint as low as 133 ms, resume as low as 49 ms

[2605.22781](#); [2605.10913](#); [2604.28138](#); [2609.35366](#); [2608.22928](#); [Kimi K3 §5.3.2](#).

Arp2/3 shapes network dynamics

In actomyosin simulations, the effect of Arp2/3 depends on concentration.



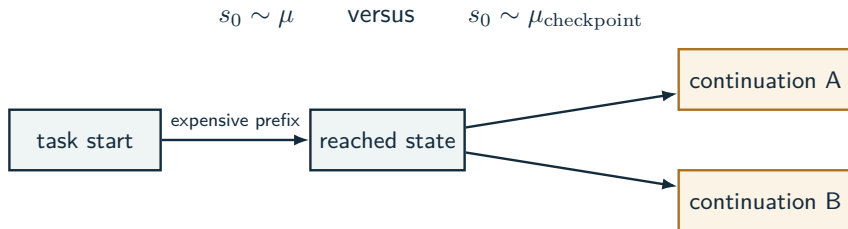
Arp2/3 regime	Reported network response
High concentration	Network stalls
Low concentration	Network contracts
Intermediate	Loosely connected clusters can collapse in motor-driven avalanches

Connectivity and interaction matter alongside branch count.

Li et al. (JPCB, 2021); Eliaz et al. (PRE, 2020); Stockmayer (JCP, 1943).

Reset changes the learning problem

The distribution of starting states determines which experience is collected.

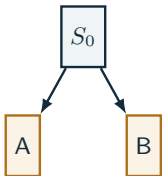


- Record the origin and selection rule for each checkpoint.
- Evaluate the final policy on the intended task distribution.

Related return-and-explore strategy: [Ecoffet et al., Go-Explore, Nature \(2021\)](#).

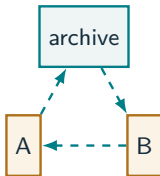
Three graphs describe the computation

Execution ancestry



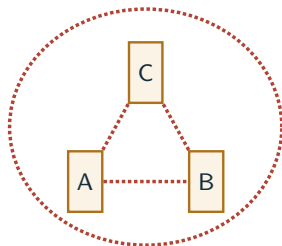
Which state produced this continuation?

Evidence flow



Which observations influenced a decision?

Composition



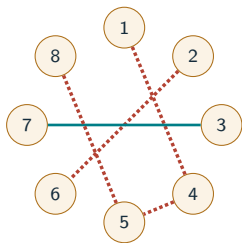
Which candidate sets need joint validation?

A runtime tree can carry communication and higher-order interactions.

How eight changes work together

Individual success does not establish the quality of a composition.

$$\Gamma_{ij} = U(\{i, j\}) - U(\{i\}) - U(\{j\}) + U(\emptyset)$$



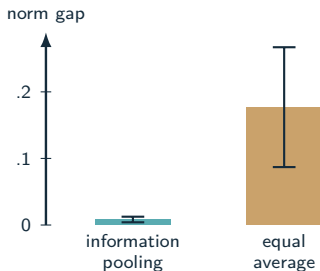
Illustrative interaction graph

- $U(A)$: fixed objective for change set A from the same parent.
- Eight candidates yield 28 pair tests.
- Higher-order effects can survive pairwise screening.
- Rebuild and evaluate each chosen composition.

What the reducer experiments establish

Unequal logistic shards

Distance to the centralized MLE; lower is closer.



0.0083 ± 0.0042 vs. 0.177 ± 0.090

Mean $\pm$ SD over eight fixed seeds.

Sample-size weighting not yet compared (paper §4). Reported synthetic results: [Eliaz \(2026\)](#), [arXiv:2607.09689](#). Error bars show SD, not confidence intervals.

Scope of the checks

- Gaussian summaries match the stated pooling algebra.
- Information weighting helps in this synthetic comparison.
- Forged precision exposes a trust failure; a heuristic is not a robustness theorem.

Ablations that explain the outcome

Vary	Observe	Question
Prefix cost and dirty pages	Resource use and memory	When does reuse pay?
No sharing / archive / routing	Quality at equal full cost	When does communication help?
Evidence overlap and coupling	Calibration and error	Does the reducer overcount?
Evaluation budget	False acceptance and time to target	Where should compute go?
Composition order and size	Joint success and regressions	Which interactions matter?

Separate a runtime gain from a search-policy or evaluation advantage.

Evaluation under adaptive search

A protected evaluator can still teach the search how to overfit its feedback.



- Keep evaluation code and admission records outside worker write access.
- Track queries and selection history.
- Use fresh validation or a justified adaptive evaluation procedure.

Russo & Zou, Controlling Bias in Adaptive Data Analysis Using Information Theory, AISTATS (2016).

The fork contract, before the fork

A future native-fork executor must satisfy all of these before enabling parallel branches:

1. fork from an immutable, content-addressed parent state
2. give every branch its own lease, TTL, budget and provenance
3. never copy publishing credentials into a coding cell
4. record the parent checkpoint, branch ID and evidence IDs for every child
5. **never count siblings that reuse the same evidence as independent agreement**
6. merge deterministically and surface conflicts as evidence, not silent retries
7. run the complete verification and review stream on the merged result
8. tear down every child on success, failure, cancellation or scheduler recovery

swfactory [docs/lifecycle.md](https://docs.swfactory.io/lifecycle.md).

TLA+ promotion invariants

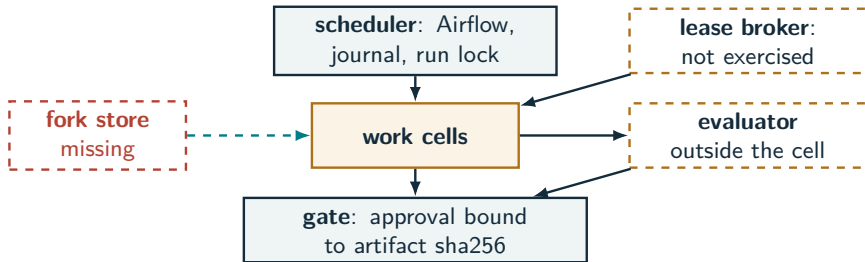
AuthorityKernel.tla models one cell across epoch replacement, Airflow binding, evidence, approval and publication.

- evidence refers only to the frozen candidate
- approval requires evidence for that same candidate
- publication requires frozen candidate = evidence = approval under the current Airflow-bound epoch
- an external commit belongs to the current epoch
- a search-origin authority request cannot become an authority source
- an `in_doubt` effect has no blind retry and leaves that state only by observation

A design model, not yet checked against traces.

swfactory [formal/authority/README.md](https://github.com/swfactory/formal/authority/README.md).

Machines fork; authority stays outside



Fork contract, 8 conditions (docs/lifecycle.md). The one that matters most here:

5: never count siblings that reuse the same evidence as independent agreement.

I wrote the fork contract before I had the fork.

swfactory docs/lifecycle.md, docs/run-recovery.md.

Fork is old, and much of it was built on Xen

	System	What it forks or moves	Reported
2003	Xen (Cambridge)	paravirtualised VMs	base of the Xen rows below
2005	Live migration (Cambridge)	pre-copy of memory to another host	60 ms downtime, Quake 3 server
2005	Potemkin	flash-cloned honeypot VMs, CoW memory	521 ms clone to first reply
2009	SnowFlock	VM_fork(N) across hosts, on Xen	600–800 ms; first example forks an untrusted child
2020	Catalyzer	sfork of a running gVisor sandbox	under 1 ms, best case
2023	Nephele	fork() for unikernel VMs, on Xen	—
2026	DeltaBox; Shepherd	agent sandbox checkpoint; fork	10.83 ms checkpoint; 134–143 ms fork

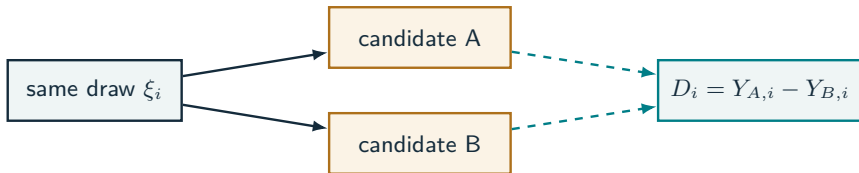
Fork lineage, 2003–2026.

SOSP'03; NSDI'05; SOSP'05; EuroSys'09; ASPLOS'20; EuroSys'23; [DeltaBox](#); [Shepherd](#).

Fork is also how you run the control

For a comparison, positive covariance helps.

$$\text{Var}(Y_A - Y_B) = \text{Var}(Y_A) + \text{Var}(Y_B) - 2 \text{Cov}(Y_A, Y_B)$$



Share randomness to compare; vary it to corroborate.

The seed policy belongs in the fork API.

Glasserman & Yao, Management Science (1992).

Research record: biophysics

PhD, University of Houston, with the Rice Center for Theoretical Biological Physics

Eliaz, Nedelec, Morrison, Levine & Cheung. Insights from graph theory on the morphologies of actomyosin networks with multilinkers. *Phys. Rev. E* 102:062420 (2020). First author.

Liman, . . . , **Eliaz**, . . . , Cheung. The role of the Arp2/3 complex in shaping the dynamics and structures of branched actomyosin networks. *PNAS* 117:10825 (2020).

Li, Liman, **Eliaz** & Cheung. Forecasting avalanches in branched actomyosin networks with network science and machine learning. *J. Phys. Chem. B* 125:11591 (2021).

Zegarra, . . . , **Eliaz**, Impact of hydrodynamic interactions on protein folding rates depends on temperature. *Phys. Rev. E* 97:032402 (2018).

Zhang, Nde, **Eliaz**, Jennings, Cieplak & Cheung. CaXML: Chemistry-informed machine learning explains mutual changes between protein conformations and calcium ions in calcium-binding proteins using structural and topological features. *Protein Science* 34:e70023 (2025).

Thesis

Eliaz. Physics of self-assembly in complex matter. PhD thesis, University of Houston (2020); self-published in book form as *Physics of Complex (Biological) Matter* (2023).

Full author lists and positions: CLAIM-FENCE.md. Meeting abstracts (Biophysical Society, APS March Meeting 2018–2021) are not listed as papers.

Research record: genomics

Genome architecture and bioinformatics

Harris, . . . , **Eliaz**, . . . , Lieberman Aiden, Rowley. Chromatin alternates between A and B compartments at kilobase scale for subgenic organization. *Nat. Commun.* 14:3303 (2023).

Kaushal, . . . , **Eliaz**, CTCF loss has limited effects on global genome architecture in *Drosophila* despite critical regulatory functions. *Nat. Commun.* 12:1011 (2021).

Barkai, . . . , **Eliaz**, OffRisk: a docker image for annotating CRISPR off-target sites in the human genome. *Bioinformatics Advances* 3:vbad138 (2023).

Cancer genomics

Saurty-Seerunghen, El-Habr, **Eliaz**, . . . , Junier. A unique malignant cell type per patient tumor encoded in each cancer cell transcriptome. *iScience* 29:115139 (2026).

Gilad, **Eliaz**, A genome-scale CRISPR Cas9 dropout screen identifies synthetically lethal targets in SRC-3 inhibited cancer cells. *Commun. Biol.* 4:399 (2021).

Gilad, **Eliaz**, Reply to: Target expression is a relevant factor in synthetic lethal screens. *Commun. Biol.* 5:836 (2022).

Gilad, **Eliaz**, Drug-induced PD-L1 expression and cell stress response in breast cancer cells can be balanced by drug combination. *Sci. Rep.* 9:15099 (2019).

Full author lists and positions: CLAIM-FENCE.md.

Research record: preprints, systems and talks

Preprints

Eliaz. Evidence-Aware MapReduce for Forkable Compute. arXiv:2607.09689 (2026). Sole author; code github.com/zozo123/boltzmann-mapreduce. Blackburn, Huber, **Eliaz**, . . . , Lieberman Aiden. Cooperation among an anonymous group protected Bitcoin during failures of decentralization. arXiv:2206.02871 (2022). **Eliaz**, Danovich & Gasic. Poolkeh finds the optimal pooling strategy for a population-wide COVID-19 testing. medRxiv (2020). First author. Hitz, . . . , **Eliaz**, . . . , Cherry. The ENCODE uniform analysis pipelines. bioRxiv (2023). The ENCODE Project Consortium (member). The Encyclopedia of DNA Elements. bioRxiv (2026). Chamo & **Eliaz**. Continuation geometry resolves apparent branch splitting in unequal-mass three-body orbits. ai.viXra:2608.0069 (2026); not peer reviewed.

Systems and open source

swfactory, an Apache Airflow software factory: github.com/zozo123/ariflow-swfactory. apache/airflow: sandbox toolset with Docker sbx and OpenSandbox backends (merged, 2026); Islo and Boat backends (open). openclaw/crabbox, nushell, and sandbox PRs to Spark, Kafka Connect and mecatl (2026). The Sandbox Shift: a field manual for running untrusted code (2026).

Talks, October 2026

Cambridge SRG (this talk), 15 Oct.
RustChinaConf, Shenzhen: Disposable Runners, Warm Cargo Factory, 17 Oct.
Zenity AI Agent Security Summit, New York: Your Agent Escaped Without Escaping the Sandbox, 21 Oct.

References: systems

Fork, snapshot, isolation

Barham et al. Xen and the art of virtualization. SOSP 2003.
 Vrabie et al. Scalability, fidelity, and containment in the Potemkin virtual honeyfarm. SOSP 2005.
 Lagar-Cavilla et al. SnowFlock. EuroSys 2009.
 Madhavapeddy et al. Jitsu. NSDI 2015.
 Manco et al. My VM is lighter (and safer) than your container. SOSP 2017.
 Baumann et al. A fork() in the road. HotOS 2019.
 Agache et al. Firecracker. NSDI 2020.
 Du et al. Catalyzer. ASPLOS 2020.
 Brooker et al. Restoring uniqueness in microVM snapshots. arXiv:2102.12892, 2021.
 Ustiugov et al. Benchmarking, analysis, and optimization of serverless function snapshots (REAP). ASPLOS 2021.
 Lupu et al. Nephele: cloning unikernel-based VMs. EuroSys 2023.
 Clark et al. Live migration of virtual machines. NSDI 2005.

Leases and fencing

Gray & Cheriton. Leases. SOSP 1989.
 Burrows. The Chubby lock service. OSDI 2006.
 Kleppmann. How to do distributed locking. 2016.

Agent sandboxes, 2026

Dong et al. DeltaBox. arXiv:2605.22781. Yu et al. Shepherd. arXiv:2605.10913. Wu et al. Crab. arXiv:2604.28138.
 Guo et al. Planarian: managing agent state with statepoints. arXiv:2609.35366. Zheng et al. arXiv:2608.22928.
 Kimi Team (Moonshot AI). Kimi K3: Open Frontier Intelligence. Tech. report, 2026, §4.2.6, §5.3.2. DeepSeek-AI. DeepSeek-V4 (DSeC). arXiv:2606.19348.
 Wei et al. No provisioned concurrency: fast RDMA-codesigned remote fork (MITOSIS). OSDI 2023. Ao et al. FaaSnap. EuroSys 2022.

Effects, authority, provenance

Nightingale, Chen & Flinn. Speculative execution in a distributed file system. SOSP 2005.
 Nightingale et al. Rethink the sync. OSDI 2006.
 Cully et al. Remus. NSDI 2008.
 Watson et al. Capsicum. USENIX Security 2010.
 Watson et al. CHERI. IEEE S&P 2015.
 Birgisson et al. Macaroons. NDSS 2014.
 Muniswamy-Reddy et al. Provenance-aware storage systems. USENIX ATC 2006.
 Pasquier et al. Practical whole-system provenance capture. SoCC 2017.
 Torres-Arias et al. in-toto. USENIX Security 2019.
 O'Callahan et al. Engineering record and replay for deployability. USENIX ATC 2017.

References: evidence and evaluation

Dependent evidence

Cochran. The combination of estimates from different experiments. *Biometrics* 1954.

Kish. *Survey Sampling*. Wiley, 1965.

Liang & Zeger. Longitudinal data analysis using generalized linear models. *Biometrika* 1986.

Hedges, Tipton & Johnson. Robust variance estimation in meta-regression with dependent effect size estimates. *Research Synthesis Methods* 2010.

Green, Karvounarakis & Tannen. Provenance semirings. *PODS* 2007.

Cameron, Gelbach & Miller. Bootstrap-based improvements for inference with clustered errors. *REStat* 2008.

Cameron, Gelbach & Miller. Robust inference with multiway clustering. *JBES* 2011.

Glasserman & Yao. Some guidelines and guarantees for common random numbers. *Management Science* 1992.

Stockmayer. Theory of molecular size distribution and gel formation in branched-chain polymers. *J. Chem. Phys.* 11:45, 1943.

Kim et al. Correlated errors in large language models. *ICML* 2025.

Evaluation under search

Dean & Barroso. The tail at scale. *CACM* 2013.

Dean. Designs, lessons and advice from building large distributed systems (“Numbers everyone should know”). *LADIS* keynote, 2009.

Brown et al. Large Language Monkeys: scaling inference compute with repeated sampling. *arXiv:2407.21787*, 2024.

Von Arx, Chan & Barnes (METR). Recent frontier models are reward hacking. 5 June 2025.

Zhong, Raghunathan & Carlini. ImpossibleBench. *arXiv:2510.20270*, 2025.

Dwork et al. The reusable holdout. *Science* 2015.

Stroebl, Kapoor & Narayanan. The limits of inference scaling through resampling (v1: Inference scaling fLaws). *arXiv:2411.17501*, 2024.

Russo & Zou. Controlling bias in adaptive data analysis using information theory. *AISTATS* 2016.

References: own work and documentation

Speaker and co-authors

Y. Eliaz. Evidence-Aware MapReduce for Forkable Compute. arXiv:2607.09689, 2026.

github.com/zozo123/ariflow-swfactory, runs SELFHOST-2/3, 27 Sep 2026.

Blackburn, Huber, Eliaz et al. Cooperation among an anonymous group protected Bitcoin during failures of decentralization. arXiv:2206.02871, 2022.

Saurty-Seerunghen, El-Habr, Eliaz et al. A unique malignant cell type per patient tumor encoded in each cancer cell transcriptome. iScience 29:115139, 2026.

Hitz, . . . , Eliaz, . . . , Cherry. The ENCODE uniform analysis pipelines. bioRxiv 2023, doi:10.1101/2023.04.04.535623.

Speaker's physics work (slide 30)

Eliaz et al. PRE 102:062420, 2020; Liman et al. PNAS 117:10825, 2020; Li et al. JPCB 125:11591, 2021.

Documentation

OBuilder (github.com/ocurrent/obuilder); day10 (tunbury.org, 16 Feb 2026); Docker Sandboxes security; Firecracker snapshotting, snapshot-support and prod-host-setup docs; CRIU TCP_REPAIR; gVisor checkpoint/restore.

Backup material

Ecoffet et al. Go-Explore. Nature 2021.